

Class-1

Question and answers:

1. Find the Fourier Transform of $f(t) = \begin{cases} 1-t, & 0 < t < 1 \\ 1+t, & -1 < t < 0. \\ 0, & |t| > 1 \end{cases}$

$$\begin{aligned} \text{Ans: } F[f(t)] &= \int_{-1}^1 f(t) e^{-i\omega t} dt \\ &= \int_{-1}^0 (1+t) e^{-i\omega t} dt + \int_0^1 (1-t) e^{-i\omega t} dt \\ &= \left[(1+t) \frac{e^{-i\omega t}}{-i\omega} - \frac{e^{-i\omega t}}{(-i\omega)^2} \right]_{-1}^0 + \left[(1-t) \frac{e^{-i\omega t}}{-i\omega} - \frac{(-1)e^{-i\omega t}}{(-i\omega)^2} \right]_0^1 \\ &= \frac{-1}{i\omega} [1-0] + \frac{1}{\omega^2} [1-e^{i\omega}] - \frac{1}{i\omega} [0-1] - \frac{1}{\omega^2} [e^{-i\omega} - 1] \\ &= \frac{1}{\omega^2} [1-e^{i\omega}] + \frac{1}{\omega^2} [1-e^{-i\omega}] \\ &= \frac{1}{\omega^2} [1-e^{i\omega} + 1-e^{-i\omega}] = \frac{2-(e^{i\omega} + e^{-i\omega})}{\omega^2} \\ &= \frac{2[1-\cos \omega]}{\omega^2}. \end{aligned}$$

2. Find the Inverse Fourier Transform of the function $\frac{e^{-2i\omega}}{(2+3i\omega)}$

$$\text{Ans: } \frac{e^{-2i\omega}}{(2+3i\omega)} = \frac{1}{3} \left(\frac{e^{-2i\omega}}{(\frac{2}{3} + i\omega)} \right)$$

$$F^{-1} \left[\frac{1}{(\frac{2}{3} + i\omega)} \right] = e^{-\frac{2}{3}t} U(t) = f(t)$$

$$F^{-1} \left[\frac{e^{-2i\omega}}{(\frac{2}{3} + i\omega)} \right] = f(t-2) = e^{-\frac{2}{3}(t-2)} U(t-2)$$

$$F^{-1} \left[\frac{e^{-2i\omega}}{(2+3i\omega)} \right] = \frac{1}{3} e^{-\frac{2}{3}(t-2)} U(t-2).$$